XG(S)PON Module Reference Manual

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Document Revision History

Revision	Date	Description
1.0	31 Oct 2023	Initial Release

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1 Document General Conventions

1.1 Overview

This manual's main purpose is to provide instructions on using the XGPON code framework and APIs. Its aim is to assist users in analyzing and clarifying their thoughts when they encounter problems. For common issues, users can easily find direct solutions or accurately provide feedback to the development team, including necessary information such as messages and logs, which will help in further analysis.

1.2 Abbreviations of Registers

List of common abbreviations in the entire reference manual:

Abbreviation	Full Name
RW	Read-Write
RO	Read-Only
WO	Write-Only
RC	Read to Clear
WC	Write to Clear
RWC	Read or Write to Clear

1.3 Abbreviations of Modules (Optional)

List common terms or abbreviations used in the reference manual. For example:

Abbreviation	Full Name
PLOAM	Physical layer operations, administration and maintenance
ONU-ID	ONU identifier
Alloc-ID	Allocation identifier
T-CONT	Transmission Container
XGEM	10-Gigabit-capable passive optical network Encapsulation Method
XGPON	10-Gigabit-capable Passive Optical Network

1. **PLOAM: The Physical Layer Operations, Administration, and Maintenance** protocol is a protocol used for physical layer operations, management, and maintenance of optical fiber access networks. It monitors and controls optical fiber transmission equipment to diagnose network faults, configure settings, perform monitoring tasks, and carry out other functions. Messages are sent and received through optical ports to

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facilitate interaction and management between communication devices.

Table 11-1 – Generic PLOAM message structure

Octet	Field	Description
1-2	ONU-ID	Ten bits, aligned at the LSB end of the 2-byte field. The six most significant bits are reserved, and should be set to 0 by the transmitter and ignored by the receiver.
3	Message type ID	This byte indicates the message type. The enumerated code point for each message type is defined below.
4	SeqNo	Sequence number.
5-40	Message_Content	The message content is defined in the clause that describes each message type ID.
41-48	MIC	Message integrity check.

1 PLOAM message format

2. **ONU-ID**: The ONU-ID in 10G GPON is a 10-bit identifier, and each ONU ID is guaranteed to be unique. In the activation phase, the OLT assigns the ONU-ID to the ONU using a PLOAM message. The validity of the ONU-ID persists continuously, with the exception of situations where power is disconnected, the OLT deactivates it, or the ONU voluntarily transitions into the Inactive state.ONU IDs have a valid range of 0 to 1023. Within this range, the numbers from 0 to 1022 represent ordinary ONU IDs that can be assigned by the OLT to individual ONUs. The ID number 1023 is reserved for downstream broadcast and unallocated ONU IDs in the upstream direction.

Octet	Content	Description
1-2	0x03FF	Broadcast ONU-ID.
3	0x03	Message type ID "Assign_ONU-ID".
4	SeqNo	Broadcast PLOAM sequence number.
5-6	ONU-ID	LSB-justified 10-bit assigned ONU-ID value padded with 6 MSB zeros; range 0-1022 (0x0000 – 0x03FE).
7-10	Vendor-ID	ONU Vendor-ID code, a four-character combination discovered at SN acquisition.
11-14	VSSN	Vendor-specific serial number, a four-byte unsigned integer discovered at SN acquisition.
15-40	Padding	Set to 0x00 by the transmitter; treated as "don't care" by the receiver.
41-48	MIC	Message integrity check.

2 ONU-ID PLOAM message format

3. **Alloc-ID:** The Alloc-ID, which is a 14-bit number, defines a traffic entity and is used for allocating upstream bandwidth. Multiple Alloc-IDs can be assigned to an ONU in XG-PON. The Assign Alloc-ID PLOAM message is used to assign Alloc-IDs to the ONU. According to the analysis of the PLOAM message format, the valid range for Alloc-IDs is from 0 to 16383. Alloc-IDs cannot be used within the range of 0 to 1022 because these values are already allocated for ONU-IDs. The value 1023 has a special meaning and should not be used as

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an Alloc-ID. The OLT can assign Alloc-IDs to ONUs within the range of 1024 to 16383.

Octet	Content	Description
1-2	ONU-ID	Directed message to one ONU.
3	0x0A	Message type ID "Assign_Alloc-ID".
4	SeqNo	Unicast PLOAM sequence number.
5-6	Alloc-ID-value	14 bits, aligned to the least significant end. The most significant bits are set to 0 by the transmitter and treated as "don't care" by the receiver. 00PPPPPP, PPPPPPPP
7	Alloc-ID-type	1-XGEM-encapsulated payload. 255-Deallocate this Alloc-ID. Other values reserved.
8-40	Padding	Set to 0x00 by the transmitter; treated as "don't care" by the receiver.
41-48	MIC	Message integrity check.

3 Assign_Alloc-ID PLOAM message format

4. **T-CONT:** T-CONT and Alloc-ID have a one-to-one mapping relationship, which is established using the OMCI protocol. One Alloc-ID can be assigned to one T-CONT, and if a second mapping attempt is made, it will be rejected by the ONU. The PLOAM protocol is responsible for creating and managing Alloc-ID, while T-CONT is created and managed by the ONU using the OMCI protocol. The creation and deletion of Alloc-IDs are handled by the PON MAC driver. Additionally, for the AIROHA chip, the concept of a channel is introduced. This channel facilitates internal hardware data transfer. There is a one-to-one mapping relationship between the channel, T-CONT, and Alloc-ID.

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2 Module (or function) name

2.1 Overview

The latest generation of broadband passive optical access standard is the 10-Gigabit Passive Optical Network (XGPON), also referred to as 10-Gigabit-Capable PON. It is based on ITU-T G.987.x specifications. XGPON, which is built upon GPON, provides higher bandwidth, improved efficiency, broader coverage, enhanced data security, and more advanced user interfaces. Most operators widely consider it as an ideal technology for achieving broadband and comprehensive transformation of access network services. The module primarily implements the underlying functions of XGPON, including ONU onboarding management, service configuration management, interrupt handling, data encryption, and other aspects.

2.2 Abbreviations

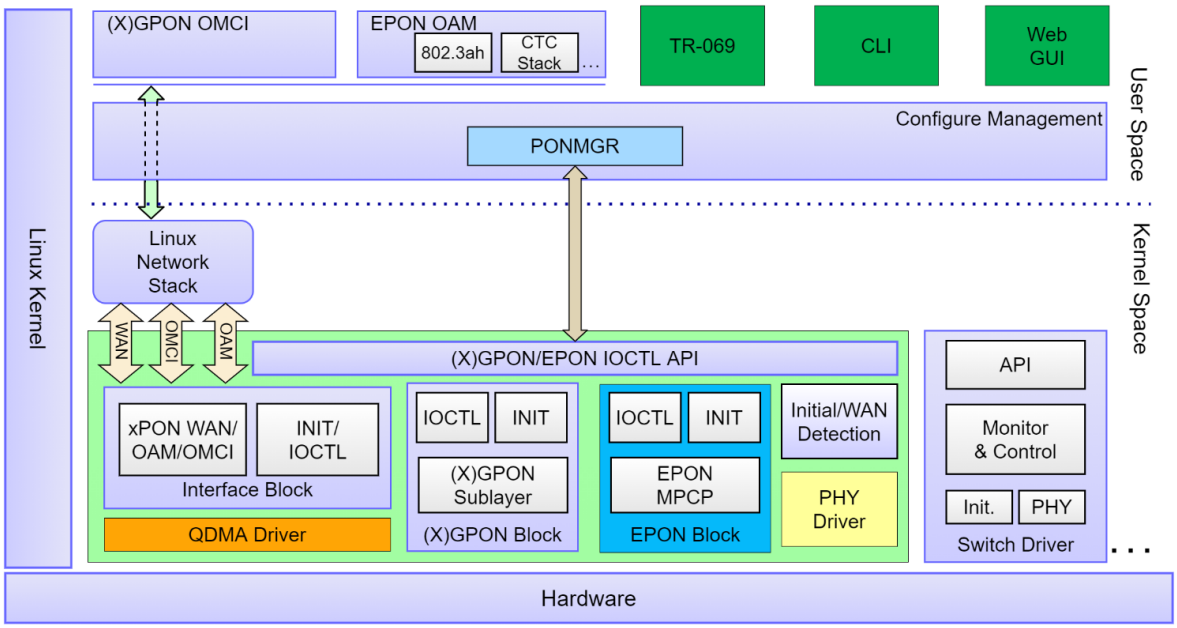
List all abbreviations used in descriptions such as block diagrams, flowcharts, tables, registers, etc.

2.3 Functional features

Describe detailed functionality.
Corresponds to the previous template 2.7 (Feature/IOT List) chapter.

2.4 Framework diagram

The frame design of XG(S)PON is essentially identical to that of GPON.



4 ONU PON Driver structure diagram

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2.5 Function

The activation process of X(S)GPON should be implemented according to the specifications G.987.X (G.987.3/G.9807.1). During the activation process, there are six states defined for the ONU:

O1 - Initial state

O2-3 - Serial_Number state

O4 - Ranging state

O5 - Operation state

O6 - Intermittent LODS state

O7 - Emergency Stop state

The states O6, which represents a transient state that occurs when there is a brief interruption or loss of optical link signal. Additionally, there is O7, which is a state where the laser of the Optical Network Unit (ONU) is shut off. This state is typically set by the Optical Line Terminal (OLT) when the ONU is found to be interfering with other users, following rogue ONU detection.

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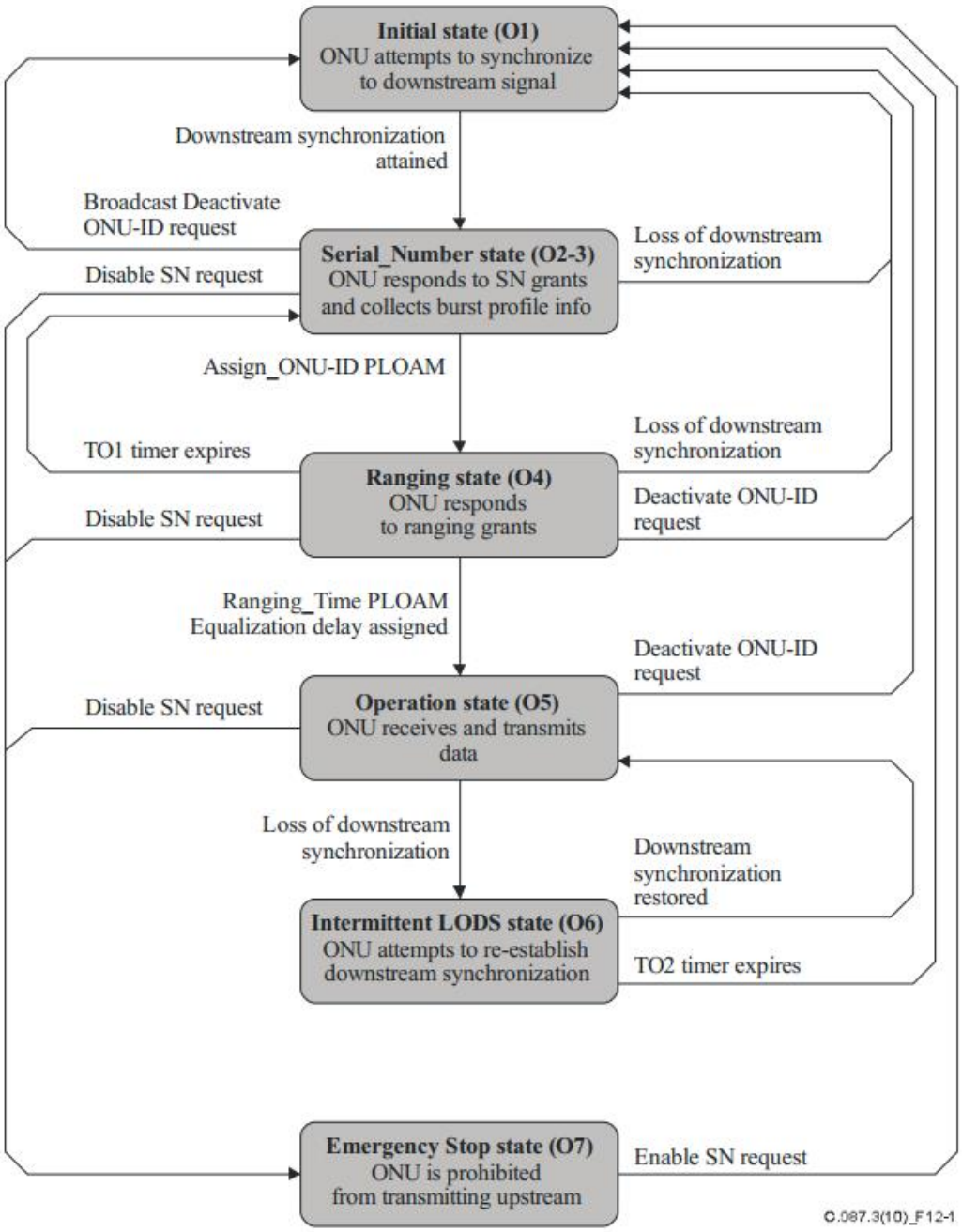

5 G.987.3 ONU activation state diagram

- Initial State(O1)

Actions	Call Functions	ONU State Change
GPON driver Initiation/Reset	gpon_dsiable ()	OX→ o1

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Actions	Call Functions	ONU State Change
Loss of downstream synchronization	gpon_phy_los_handler()	O2_3/O4->O1
Deactivate ONU-ID request	ploam_recv_deactivate_onu()	O4/O5->O1
TO2 Timer Expires	gpon_act_to2_timer_expires();	O6→O1
Receive Disable PLOAM – Enable specific SN option (SN match)	ploam_receive_disable_serial_number	O7 → O1

- Serial Number State (O2-O3)

Actions	Call Functions	ONU State Change
ONU Received Downstream Frame Clear LOS/LOF	gpon_phy_ready_handler()	O1 →O2_3
TO1 Timer Expires	gpon_act_to1_timer_expires();	O4 →O2_3

- Ranging State (O4)

Actions	Call Functions	ONU State Change
ONU Received ONU_ID	ploam_recv_assign_onu_id()	O2_3→ O4

- Operation State (O5)

Actions	Call Functions	ONU State Change
ONU Received an Equalization Delay	ploam_recv_ranging_time()	O4→ O5
Downstream synchronization restored	gpon_phy_ready_handler()	O6 → O5

- Intermittent LODS state (O6)

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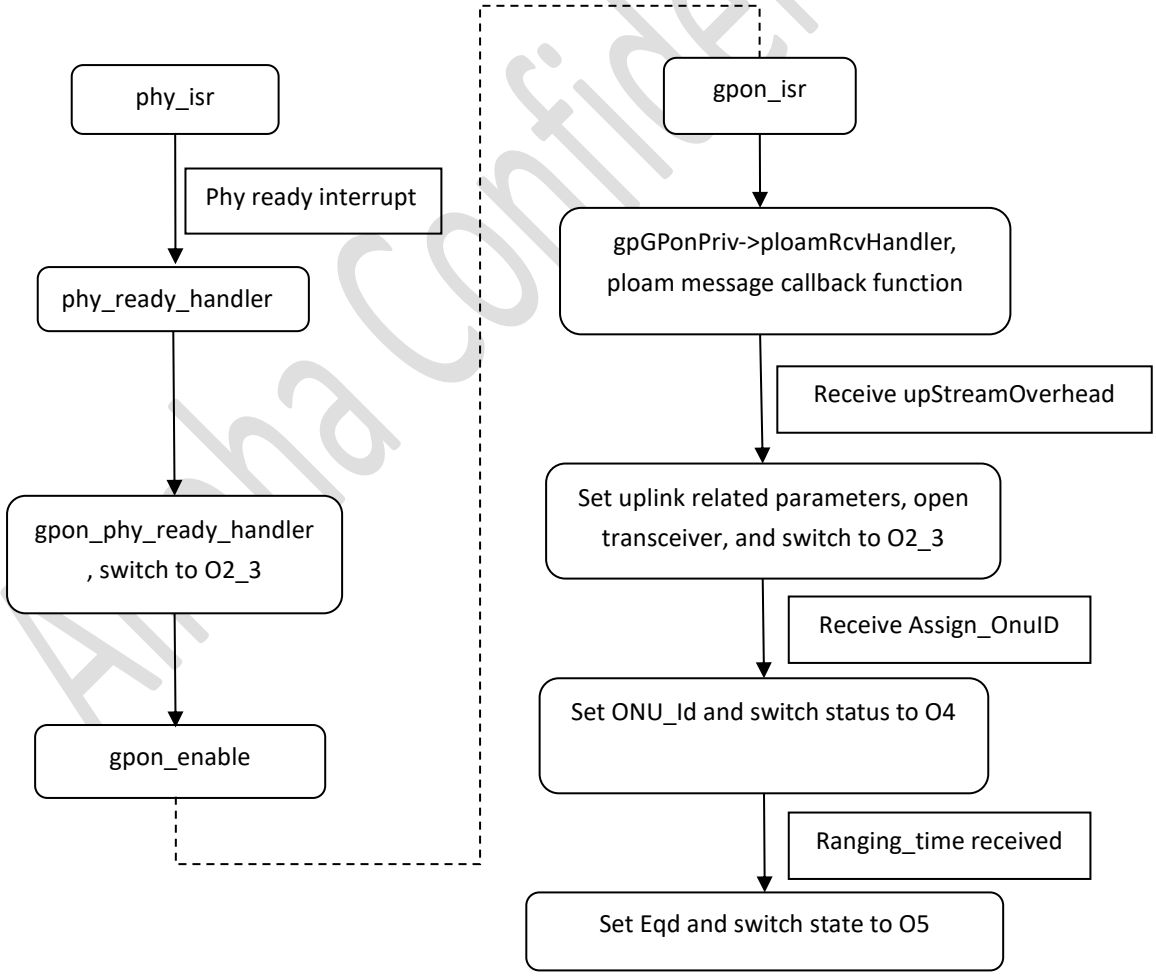
Actions	Call Functions	ONU State Change
Loss of downstream synchronization	gpon_phy_los_handler()	O5 → O6

- Emergency Stop State (O7)

Actions	Call Functions	ONU State Change
Receive Disable PLOAM – Disable specific SN option (SN match)	ploam_receive_disable_serial_number()	OX → O7
ONU Operation as Emergency State:	xmcs_set_emergency_state();	O1 → O7

The state switching associated with the registration process of ONU is implemented in the driver for XG(S)PON, as depicted in Figure 3.



6 State switching in XG(S)PON Driver

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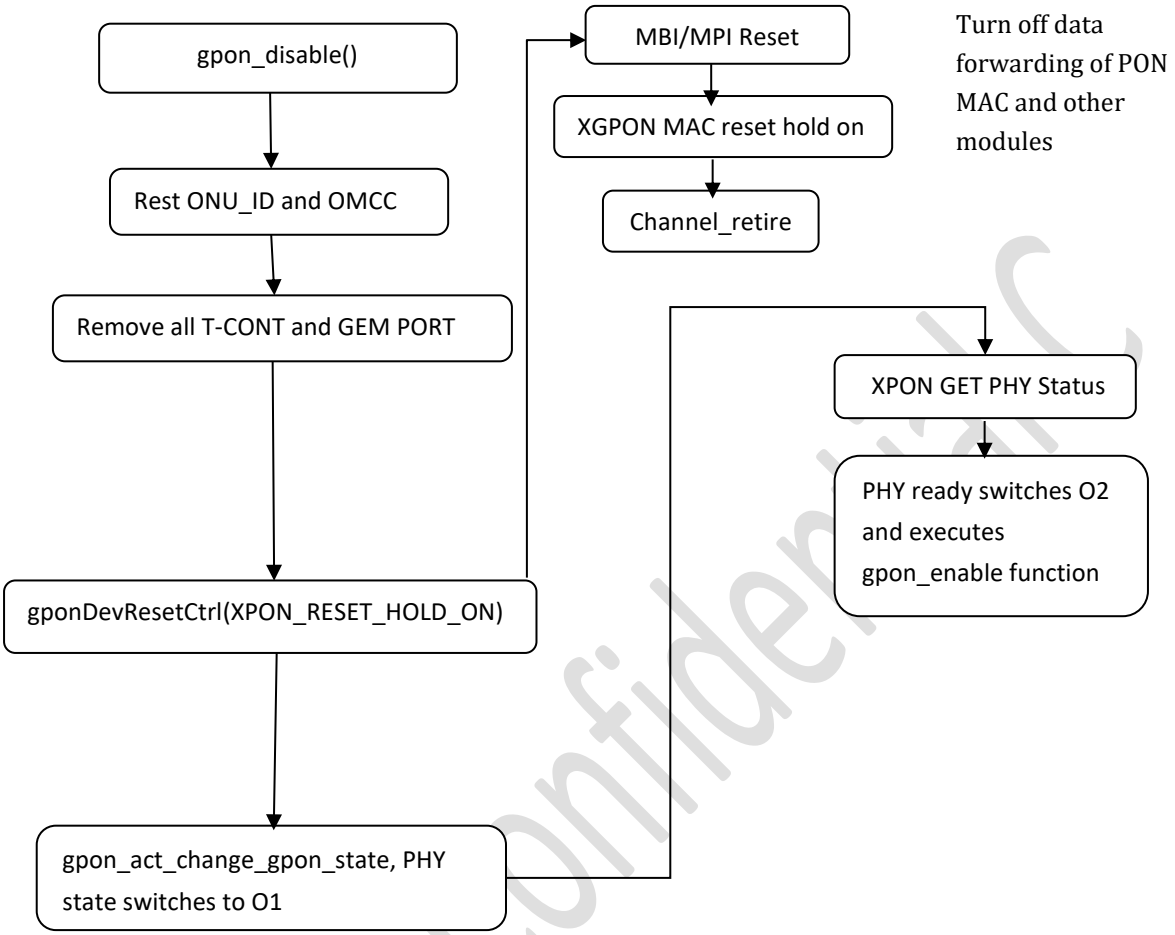
PHY los or initialization process after receiving Deactivate_ONU-ID.



7 PHY lost or received Deactivate_ONU-ID XG(S)PON MAC initialization process

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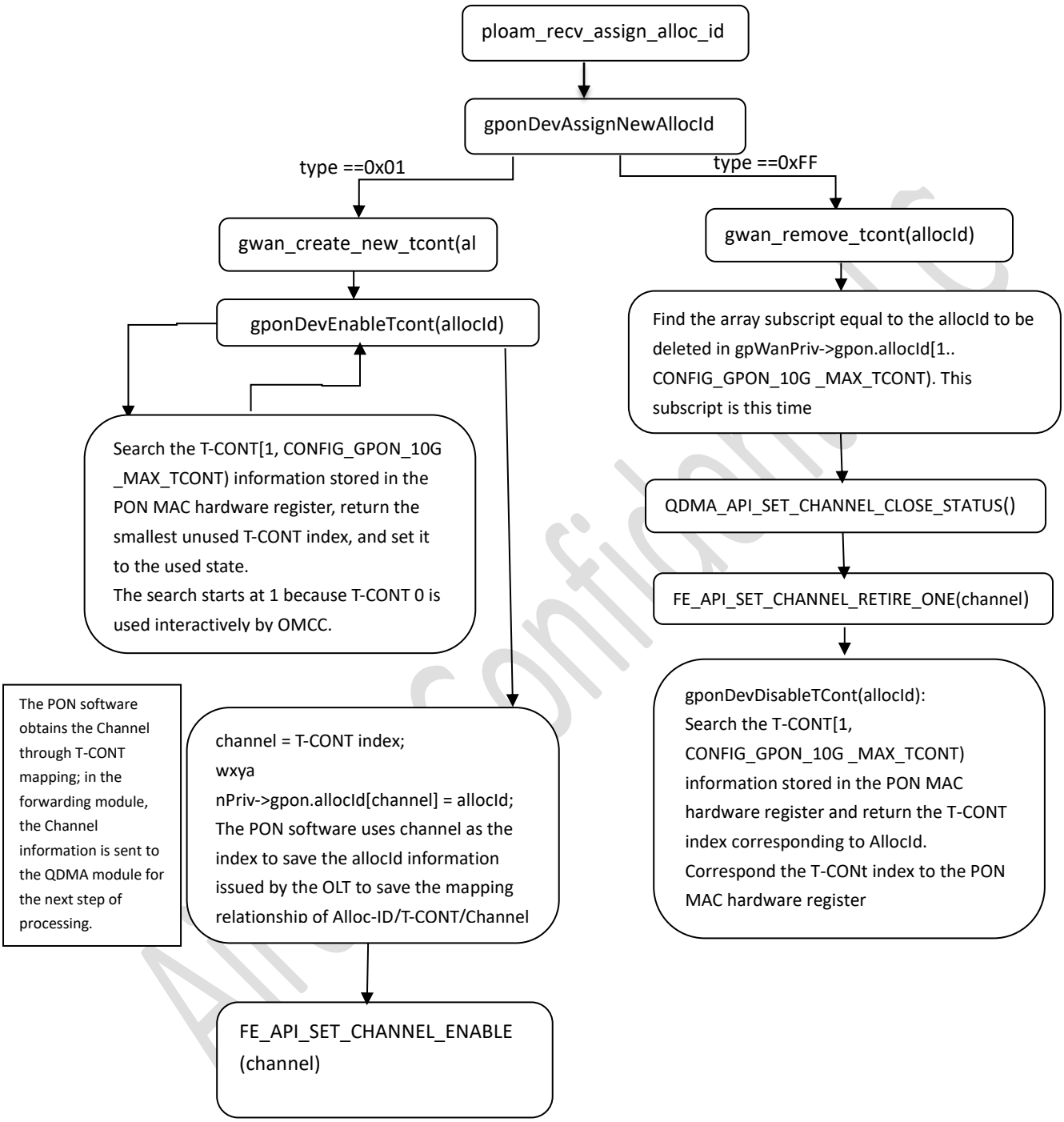

8 Processing of assign_Alloc-ID

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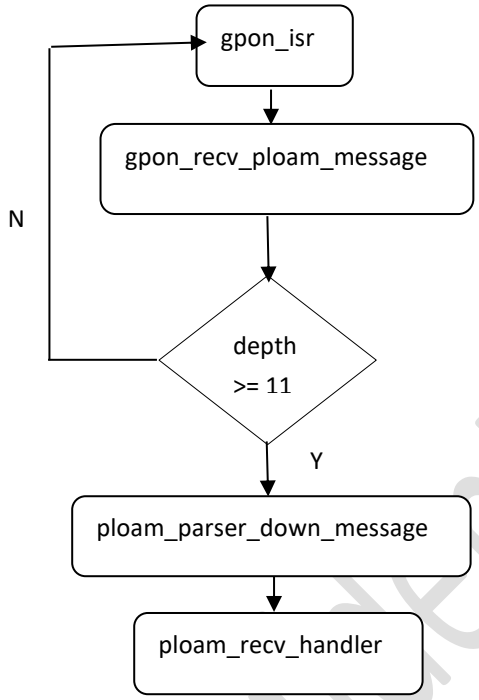
2. Processing of PLOAM messages



9 PLOAM message processing

PLOAM Message	Call Functions
PLOAM_DOWN_MSG_PROFILE	ploam_recv_profile
PLOAM_DOWN_MSG_ASSIGN_OUNID	ploam_recv_assign_onu_id
PLOAM_DOWN_MSG_RANGING_TIME	ploam_recv_ranging_time
PLOAM_DOWN_MSG_DEACTIVATE_ONUID	ploam_recv_deactivate_onu
PLOAM_DOWN_MSG_DISABLE_SERIAL_NUM	ploam_recv_disable_serial_number
PLOAM_DOWN_MSG_REQUEST_REGISTRATION	ploam_recv_requset_registration
PLOAM_DOWN_MSG_ASSIGN_ALLOCID	ploam_recv_assign_alloc_id
PLOAM_DOWN_MSG_KEY_CONTROL	ploam_recv_key_control
PLOAM_DOWN_MSG_SLEEP_ALLOW	ploam_recv_sleep_allow

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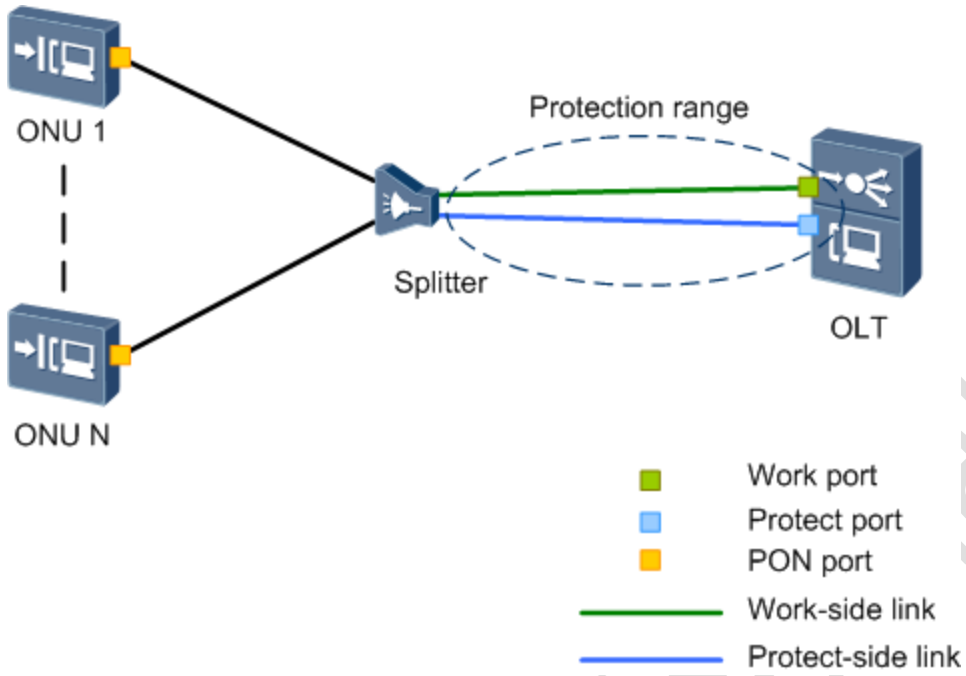
3. Introduction to TypeB switching process

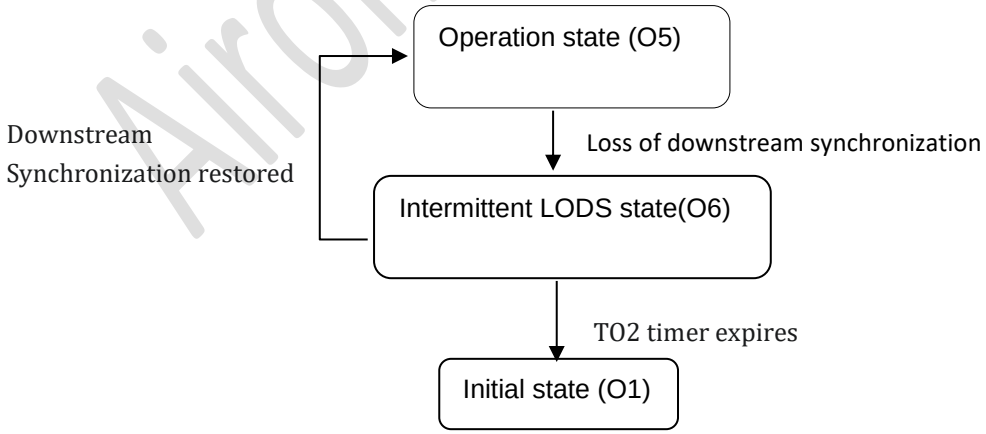


10 TYPE B topology diagram

Type B refers to Type B protect, which is a protection mechanism used in Passive Optical Networks (PON). When there is a need to switch the link of the Optical Line Terminal (OLT) PON port, it can be switched from one PON port to another.

Scenario example:

- The main fiber of the working pon is damaged.
- The olt slot of working pon needs to be upgraded.



11 TYPE B state transition diagram

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```
# [4296322527]XGPON IRQ: bwmap check error interrupt,status:0000c004
[4296322547]XGPON IRQ: bwmap check error interrupt,status:0000c004
[4296323082]XGPON IRQ: bwmap check error interrupt,status:0000c004
[4296323102]XGPON IRQ: bwmap check error interrupt,status:0000c004
<ISR> PHY_LINK_STATUS_LOS XGPON ISR=12=6+3+2+5+0,REG IntStatus=0x00400011
[4296323122]PHY IRQ: gpon detect phy_loss.
[4296323122]PLOAM: State change: original = 05  new = 06
[4296323122]Start the T2/T3 timer. Current State:06 protectFlag=1
[4296323122]xmcs_report_event begin ,type 1 id 10 value 6
[4296323122]Change the current state: 06. Last State:05
<ISR> RX_LOF XGPON ISR=13=6+3+2+6+0,REG IntStatus=0x00400014
[4296323133]T02 timer timeout. Current State:06, Next State:01
[4296323133]gpon_disable: mac plain reset.
[4296323134]MAC RESET: reset hold on
[4296323137]PLOAM: State change: original = 06  new = 01
[4296323137]xmcs_report_event begin ,type 1 id 12 value 0
[4296323137]xmcs_report_event begin ,type 1 id 10 value 1
[4296323139]Change the current state: 01. Last State:06
[4296323139]xmcs_report_event begin ,type 1 id 0 value 0
 - POLLING - __START__ [en7581_xgpon_phy_event_poll:5582] phy_status=2 (ready=0,los=1)
 | POLLING => NO laser! (1) ReadyStatus=0,LosStatus=1
 - POLLING -  __END__  (3)(1)(0) Get phy_status = 0 (3)(7)
<ISR> RX_RDY Reset__Serdes XGPON ISR=14=6+3+3+6+0,REG IntStatus=0x00000200
<ISR> PHY_LINK_STATUS_READY XGPON ISR=15=7+3+3+6+0,REG IntStatus=0x00000002
[4296323452]PHY IRQ: xgpon detect phy ready.
[4296323452]gpon_enable.
[4296323452]MAC RESET: reset release
[4296323452]PLOAM: State change: original = 01  new = 02
[4296323452]No timer start in current State:02
[4296323452]xmcs_report_event begin ,type 1 id 10 value 2
[4296323452]Change the current state: 02. Last State:01
[4296323452]xmcs_report_event begin ,type 1 id 1 value 0
killall: epon_oam: no process killed
 - POLLING - __START__ [en7581_xgpon_phy_event_poll:5582] phy_status=1 (ready=1,los=0)
 - POLLING - __END__  (3)(1)(0) Get phy_status = 1 (3)(8)
[4296323637]XGPON IRQ: SN_Request received interrupt.
[4296323637]XGPON IRQ: SN sent in O2_3 state interrupt.
[4296323657]XGPON IRQ: SN_Request received interrupt.
[4296323657]XGPON IRQ: SN sent in O2_3 state interrupt.
[4296323678]XGPON IRQ: SN_Request received interrupt.
[4296323678]XGPON IRQ: SN sent in O2_3 state interrupt.
[4296323698]XGPON IRQ: A PLOAM message received interrupt.

[4296323698]PLOAM: Receive Assign_ONU_ID message: 03FF036B 00044D54 4B472023 27120000 00000000 00000000
00000001
[4296323698]PLOAM: set ONU_ID:4 in O2_3
[4296323698]PLOAM: State change: original = 02  new = 04
[4296323698]Start the T1 timer. Current State:04
[4296323698]xmcs_report_event begin ,type 1 id 10 value 4
[4296323698]Change the current state: 04. Last State:02
[4296323698]Receive the PLOAM message same:6B03FF03
[4296323698]Receive the PLOAM message same:6B03FF03
[4296323703]XGPON IRQ: A PLOAM message received interrupt.
[4296323717]XGPON IRQ: Ranging_Request received interrupt.
[4296323717]XGPON IRQ: registration send in 04 interrupt.
[4296323719]XGPON IRQ: A PLOAM message received interrupt.

[4296323719]PLOAM: Receive Ranging Time message: 00040421 0100096F EF000000 00000000 00000000 00000000
0000001
[4296323719]PLOAM: no EqD offset in 04
[4296323719]PLOAM: set directed ranging time in 04, absolute eqd:0x00096FEF
[4296323719]PLOAM: State change: original = 04  new = 05
```

12 Online process log after normal fiber insertion and removal

5. Rogue ONU diagnostic procedure

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11.3.3.5 Disable_Serial_Number message

Octet	Content	Description
1-2	0x03FF	Broadcast ONU-ID.
3	0x06	Message type ID "Disable_Serial_Number".
4	SeqNo	Broadcast PLOAM sequence number.

Octet	Content	Description
5	Disable/enable	0xFF-The ONU with this serial number is denied upstream access. 0x00-The ONU with this serial number is allowed upstream access. 0x0F-All ONUs are denied upstream access. The content of bytes 6-13 is ignored. 0xF0-All ONUs are allowed upstream access.
6-9	Vendor-ID	ONU Vendor-ID code, a four-character combination discovered at SN acquisition.
10-13	VSSN	Vendor-specific serial number, a four-byte unsigned integer discovered at SN acquisition.
14-40	Padding	Set to 0x00 by the transmitter; treated as "don't care" by the receiver.
41-48	MIC	Message integrity check.

13 Disable_Serial_Number PLOAM message format

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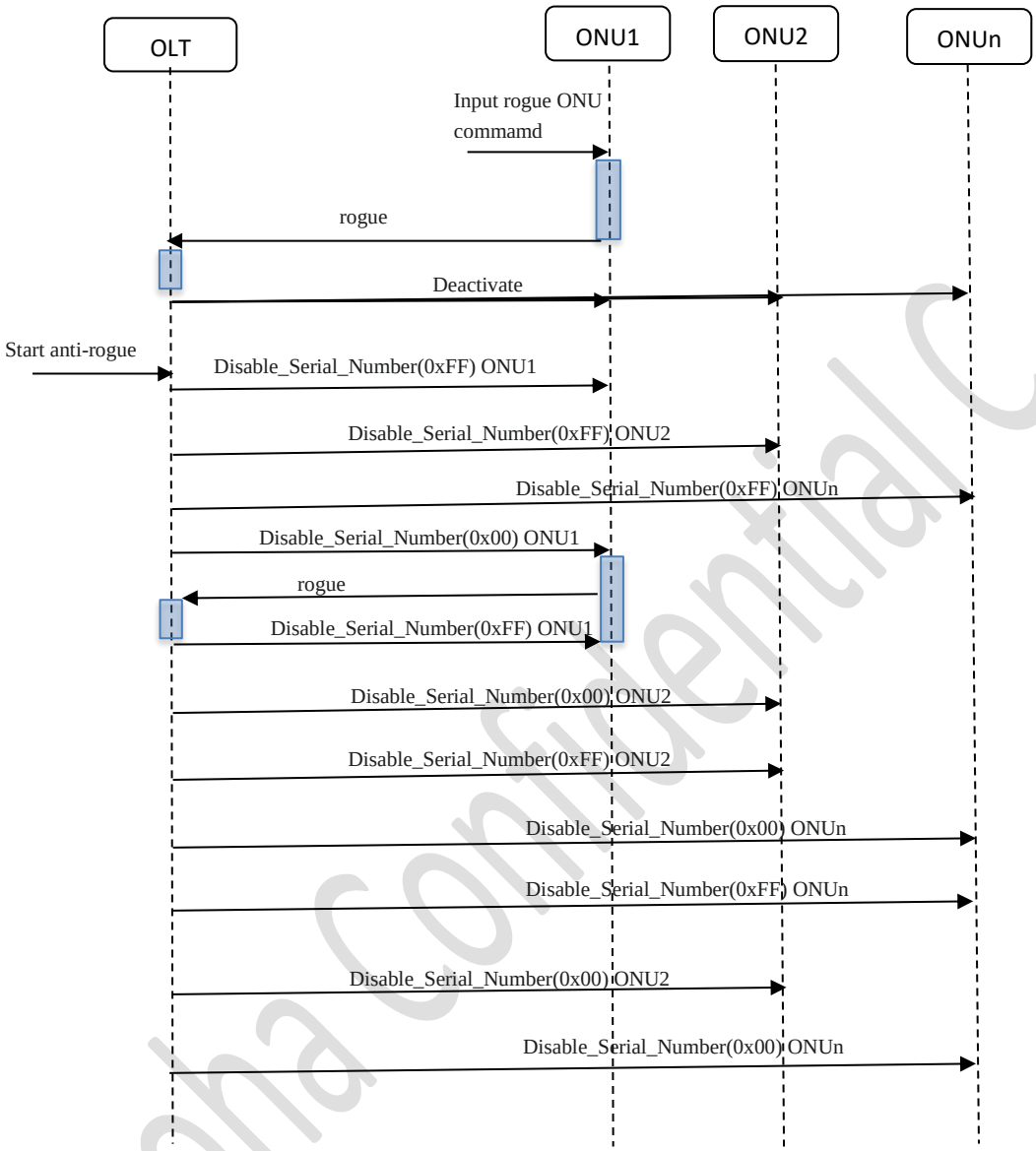

14 ***Rogue ONU detection process***

2.6 Programming Guide

```
Compile command
osbnq64 fakeroot make -f app_bsp/MakeFile_Main_BSP PROFILE=XXX CUSTOM=XX xpon_modules

The command to load modules
insmod /lib/modules/xpon.ko
```


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code pathBSP/xpon_bsp/app_bsp/
BSP/xpon_bsp/modules/xpon/

2.6.1 Get APIs

2.6.1.1 gponmgr_lib_get_omci_dbg_level

Function Prototype	**int gponmgr_lib_get_omci_debug_level(unsigned char *DebugLevel)**	
Description	Get the current debug level of OMCI.	
Support Chip	EN7521	
Parameter 1	DebugLevel	debug level of OMCI
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.2 gponmgr_lib_get_lan_debug

Function Prototype	**int gponmgr_lib_get_lan_debug(unsigned char *LanDebug)**	
Description	Get the information of LAN debug whether enabled or not.	
Support Chip	EN7521	
Parameter 1	LanDebug	LAN debug whether enabled
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.3 gponmgr_lib_get_onu_version

Function Prototype	**int gponmgr_lib_get_onu_version(char *onu_ver)**	
Description	Get the version of the ONU as defined by the vendor.	
Support Chip	EN7521	
Parameter 1	onu_ver	The version of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.4 gponmgr_lib_get_vendor_id

Function Prototype	**int gponmgr_lib_get_vendor_id(char *VendorId)**	
Description	Get to know the vendor of the ONU.	
Support Chip	EN7521	
Parameter 1	VendorId	The vendor ID of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.5 gponmgr_lib_get_equipment_id

Function Prototype	**int gponmgr_lib_get_equipment_id(char *equipmentId)**	
Description	Get the equipment ID of the ONU. It is used to identify the specific type of ONU.	
Support Chip	EN7521	
Parameter 1	equipmentId	The equipment ID of ONU

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Function Prototype	**int gponmgr_lib_get_equipment_id(char *equipmentId)**
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL

2.6.1.6 gponmgr_lib_get_operator_id_cap

Function Prototype	**int gponmgr_lib_get_operator_id_cap(char *operatorID_cap)**	
Description	Get the operator ID of ONU capability ME.	
Support Chip	EN7521	
Parameter 1	operatorID_cap	The operator ID
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.7 gponmgr_lib_get_operator_id_auth

Function Prototype	**int gponmgr_lib_get_operator_id_auth(char * operatorID_auth)**	
Description	Get the operator ID of LOID authentication ME.	
Support Chip	EN7521	
Parameter 1	operatorID_auth	The operator ID for LOID authentication
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.8 gponmgr_lib_get_loid

Function Prototype	**int gponmgr_lib_get_loid(char * Loid)**	
Description	Get the logical ONU ID.	
Support Chip	EN7521	
Parameter 1	Loid	logical ONU ID
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.9 gponmgr_lib_get_loid_password

Function Prototype	**int gponmgr_lib_get_loid_password(char * LOID_Password)**	
Description	Get logical password. It is used to provide a way for the ONU to submit authentication credentials.	
Support Chip	EN7521	
Parameter 1	LOID_Password	logical password
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.10 gponmgr_lib_get_enhanced_psk

Function Prototype	**int gponmgr_lib_get_enhanced_psk(unsigned char * enhanced_psk)**	
Description	Get an ONU enhanced PSK.	
Support Chip	EN7521	
Parameter 1	enhanced_psk	The enhanced PSK
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.11 gponmgr_lib_get_link_mode

Function Prototype	**int gponmgr_lib_get_link_mode(char * linkMode)**	
Description	Get the link mode of GPON.	
Support Chip	EN7521	
Parameter 1	linkMode	The link mode of GPON
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.12 gponmgr_lib_get_vendor_pro_code

Function Prototype	**int gponmgr_lib_get_vendor_pro_code(unsigned short * vendorProCode)**	
Description	Get the vendor specific product code for the ONU.	
Support Chip	EN7521	
Parameter 1	vendorProCode	The vendor specific product code for the ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.13 gponmgr_lib_get_omcc_version

Function Prototype	**int gponmgr_lib_get_omcc_version(unsigned char * omccVersion)**	
Description	To get the version of the OMCC protocol being used by the ONU.	
Support Chip	EN7521	
Parameter 1	omccVersion	The version of OMCC
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.14 gponmgr_lib_get_tm_option

Function Prototype	**int gponmgr_lib_get_tm_option(unsigned char * tmOption)**	
Description	To get the upstream traffic management function implemented in the ONU.	
Support Chip	EN7521	
Parameter 1	tmOption	The traffic management option of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.15 gponmgr_lib_get_survival_time

Function Prototype	**int gponmgr_lib_get_survival_time(unsigned char * onuSurvivalTime)**	
Description	To get the minimum guaranteed time in milliseconds between the loss of external power and the silence of the ONU.	
Support Chip	EN7521	
Parameter 1	onuSurvivalTime	The survival time of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.16 gponmgr_lib_get_extended_tclayer

Function Prototype	**int gponmgr_lib_get_extended_tclayer(unsigned char * extendedTcLayer)**	
Description	Get the information that whether the ONU supports or does not support various optional TC-layer capabilities of [ITU-T G.984.3].	
Support Chip	EN7521	
Parameter 1	extendedTcLayer	Bit 1(LSB): ITU-T G.984.3 Annex C, PON-ID maintenance Bit 2: ITU-T G.984.3 Annex D, PLOAM channel enhancements: swift_POPUP and ranging_adjustment messages. Bit 3..16: Reserved
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.17 gponmgr_lib_get_security_cap

Function Prototype	**int gponmgr_lib_get_security_cap(unsigned char * securityCap)**	
Description	Get the security capabilities of the ONU.	
Support Chip	EN7521	
Parameter 1	securityCap	the security capabilities of the ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.18 gponmgr_lib_get_total_ts_num

Function Prototype	**int gponmgr_lib_get_total_ts_num(unsigned char * totalTSNum)**	
Description	Get the total number of traffic schedulers that are not associated with a circuit pack, but with the ONU in its entirety.	
Support Chip	EN7521	
Parameter 1	totalTSNum	The total number of traffic schedulers
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.19 gponmgr_lib_get_total_pq_num

Function Prototype	**int gponmgr_lib_get_total_ts_num(unsigned char * totalPQNum)**	
Description	Get the total number of upstream priority queues that are not associated with a circuit pack, but with the ONU in its entirety.	
Support Chip	EN7521	
Parameter 1	totalPQNum	The total number of priority queues
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.20 gponmgr_lib_get_total_gemport_num

Function Prototype	**int gponmgr_lib_get_total_gemport_num(unsigned char * totalGEMPortNum)**
Description	Get the total number of GEM port-IDs supported by the ONU.

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Function Prototype	int gponmgr_lib_get_total_gemport_num(unsigned char * totalGEMPortNum)	
Support Chip	EN7521	
Parameter 1	totalGEMPortNum	total number of GEM ports
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.21 gponmgr_lib_get_connectivity_cap

Function Prototype	int gponmgr_lib_get_connectivity_cap(unsigned short * connectivityCap)	
Description	Get the Ethernet connectivity models that the ONU can support.	
Support Chip	EN7521	
Parameter 1	connectivityCap	the Ethernet connectivity models that the ONU can support.
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.22 gponmgr_lib_get_qos_conf_flex

Function Prototype	int gponmgr_lib_get_qos_conf_flex(unsigned short * qosConfFlex)	
Description	Get the information that whether various MEs in the ONU are fixed by ONU's architecture or whether they are configurable.	
Support Chip	EN7521	
Parameter 1	qosConfFlex	QoS configuration flexibility
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.23 gponmgr_lib_get_voip_by_tr069

Function Prototype	int gponmgr_lib_get_voip_by_tr069(char * VoIPbyTR069)	
Description	Get the VoIP configuration method whether it is TR069.	
Support Chip	EN7521	
Parameter 1	VoIPbyTR069	Whether is TR069
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.24 gponmgr_lib_get_spec_version

Function Prototype	int gponmgr_lib_get_spec_version(unsigned char * SpecVersion)	
Description	Get the spec version of ONU capability ME.	
Support Chip	EN7521	
Parameter 1	VoIPbyTR069	CTC spec version
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.25 gponmgr_lib_get_onu_power_ctrl

Function Prototype	int gponmgr_lib_get_onu_power_ctrl(unsigned char * ONU_PowerCtr)
Description	Whether the Tx power of the ONU can be controlled independently.
Support Chip	EN7521

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Function Prototype	int gponmgr_lib_get_onu_power_ctrl(unsigned char * ONU_PowerCtr)	
Parameter 1	ONU_PowerCtr	0x00:do not support ONU power control 0x01:Only ONU Tx power control is supported 0x02:ONU Tx and Rx power can be independently controlled
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.26 gponmgr_lib_get_auth_status

Function Prototype	int gponmgr_lib_get_auth_status(unsigned char * AuthStatus)	
Description	Get the credentials status of LOID authentication	
Support Chip	EN7521	
Parameter 1	AuthStatus	The credentials status of LOID authentication
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.27 gponmgr_lib_get_max_ports

Function Prototype	int gponmgr_lib_get_max_ports(char * max_ports)	
Description	Get the largest port number contained in the port list attributes.	
Support Chip	EN7521	
Parameter 1	max_ports	The maximum number of ports
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.28 gponmgr_lib_get_port_list

Function Prototype	int gponmgr_lib_get_port_list(int * port_id)	
Description	Get the port list of ONU .	
Support Chip	EN7521	
Parameter 1	port_id	The port list of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.29 gponmgr_lib_get_sr_indication

Function Prototype	int gponmgr_lib_get_sr_indication(unsigned char * srIndication)	
Description	Get the queue status for the DBA.	
Support Chip	EN7521	
Parameter 1	srIndication	The value true means that status reporting is available for all T-CONTs that are associated with the ANI.
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.30 gponmgr_lib_get_total_tcont_num

Function Prototype	int gponmgr_lib_get_total_tcont_num(unsigned short * totalTcontNum)
Description	Get the total number of T-CONTs that can be supported on this ANI.

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Function Prototype	int gponmgr_lib_get_total_tcont_num(unsigned short * totalTcontNum)	
Support Chip	EN7521	
Parameter 1	totalTcontNum	The total number of T-CONTs
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.31 gponmgr_lib_get_gem_block_length

Function Prototype	int gponmgr_lib_get_gem_block_length(unsigned short *gemBlockLength)	
Description	Get the queue occupancy reporting granularity for DBA, in units of bytes.	
Support Chip	EN7521	
Parameter 1	gemBlockLength	GEM block length
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.32 gponmgr_lib_get_onu_response_time

Function Prototype	int gponmgr_lib_get_onu_response_time(unsigned short *onuResponseTime)	
Description	Get the ONU's actual response time.	
Support Chip	EN7521	
Parameter 1	onuResponseTime	The response time of the ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.33 gponmgr_lib_get_con_req_port

Function Prototype	int gponmgr_lib_get_con_req_port(unsigned short *conReqPort)	
Description	Get the TCP or UDP port value as assigned by IANA for the management protocol associated with this virtual Ethernet interface.	
Support Chip	EN7521	
Parameter 1	conReqPort	IANA assigned port
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.34 gponmgr_lib_get_dev_mac

Function Prototype	int gponmgr_lib_get_dev_mac(unsigned char *devMac)	
Description	Get the MAC address of the ONU .	
Support Chip	EN7521	
Parameter 1	devMac	The MAC address of the ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.35 gponmgr_lib_get_gpon_state

Function Prototype	int gponmgr_lib_get_gpon_state(unsigned char *state)
Description	Get the state of the ONU.
Support Chip	EN7521

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Function Prototype	int gponmgr_lib_get_gpon_state(unsigned char *state)	
Parameter 1	state	The state of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.36 gponmgr_lib_get_onu_sn

Function Prototype	int gponmgr_lib_get_onu_sn(unsigned char *onu_sn)	
Description	To get the serial number of the ONU.	
Support Chip	EN7521	
Parameter 1	onu_sn	The serial number of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.37 gponmgr_lib_get_onu_type

Function Prototype	int gponmgr_lib_get_onu_type(int *OnuType)	
Description	Get the type of ONU.	
Support Chip	EN7521	
Parameter 1	OnuType	The type of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.38 gponmgr_lib_get_qos_scheduler

Function Prototype	int gponmgr_lib_get_qos_scheduler(unsigned char channel,struct XMCS_ChannelQoS_S *pwanChannelQos)	
Description	Get the information about the QoS scheduler.	
Support Chip	EN7521	
Parameter 1	channel	Channel
Parameter 2	pwanChannelQos	The type of QoS scheduler
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.39 gponmgr_lib_get_qos_weight_cfg

Function Prototype	int gponmgr_lib_get_qos_weight_cfg(struct XMCS_QosWeightConfig_S *pwanQosWeightCfg)	
Description	Get the type of QoS weight.	
Support Chip	EN7521	
Parameter 1	pwanQosWeightCfg	The type of QoS weight
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.40 gponmgr_lib_get_congest_config

Function Prototype	**int gponmgr_lib_get_congest_config(struct XMCS_TxQueueCongestion_S *pwanCongest)**	
Description	Get the information about the congestion.	
Support Chip	EN7521	
Parameter 1	pwanCongest	the information of congestion
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.41 gponmgr_lib_get_trtcm_scale

Function Prototype	**int gponmgr_lib_get_trtcm_scale(struct XMCS_TxTrtcmScale_S *pwanTrtcmScale)**	
Description	Get the scale of trTCM.	
Support Chip	EN7521	
Parameter 1	pwanTrtcmScale	The scale of trTCM
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.42 gponmgr_lib_get_trtcm_para

Function Prototype	**int gponmgr_lib_get_trtcm_para(unsigned char tsidx, struct XMCS_TxQueueTrtcm_S *pwanTrtcmparameters)**	
Description	Get the related parameters of trTCM,such as CIR, PIR and so on.	
Support Chip	EN7521	
Parameter 1	tsidx	index
Parameter 2	pwanTrtcmparameters	The related parameters of trTCM
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.43 gponmgr_lib_get_softimage

Function Prototype	**int gponmgr_lib_get_softimage(gpon_softimage_t * softimage,int instance)**	
Description	Get the related parameters of the software image ME.	
Support Chip	EN7521	
Parameter 1	softimage	the related parameters
Parameter 2	instance	0 or 1
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.44 gponmgr_lib_get_current_tod

Function Prototype	**int gponmgr_lib_get_current_tod(struct XMCS_GponTodCfg_S *pgponTodCfg)**	
Description	Get current ToD.	
Support Chip	EN7521	
Parameter 1	pgponTodCfg	current ToD

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Function Prototype	**int gponmgr_lib_get_current_tod(struct XMCS_GponTodCfg_S *pgponTodCfg)**	
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.45 gponmgr_lib_get_gemport_statistic

Function Prototype	**int gponmgr_lib_get_gemport_statistic(GEM_STATISTIC_t *statistic)**	
Description	Get the statistical information about GEM port.	
Support Chip	EN7521	
Parameter 1	statistic	The statistical information of GEM port
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.46 gponmgr_lib_get_wan_statistic

Function Prototype	**int gponmgr_lib_get_wan_statistic(WAN_STATISTIC_t *pWanCntStats)**	
Description	Get the statistical information of WAN.	
Support Chip	EN7521	
Parameter 1	pWanCntStats	the statistical information of WAN
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.47 gponmgr_lib_get_ploam_gtc_info

Function Prototype	**int gponmgr_lib_get_ploam_gtc_info(PLOAM_GTC_INFO_t *gtc_info)**	
Description	Get the information about PLOAM.	
Support Chip	EN7521	
Parameter 1	gtc_info	the information about PLOAM
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.48 gponmgr_lib_get_drv_power_level

Function Prototype	**int gponmgr_lib_get_drv_power_level(PLOAM_POWER_LEVEL_t *level)**	
Description	Get the ONU TX power level mode used by the ONU.	
Support Chip	EN7521	
Parameter 1	level	ONU TX power level mode
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.49 gponmgr_lib_get_all_tcont_packet_info

Function Prototype	**int gponmgr_lib_get_all_tcont_packet_info(ALL_TCONT_COUNTERS_T *aTcontCounters)**	
Description	Get the information of all TCONTs.	
Support Chip	EN7521	
Parameter 1	aTcontCounters	The information of all TCONTs
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.50 gponmgr_lib_get_tcont_packet_info

Function Prototype	**int gponmgr_lib_get_all_tcont_packet_info(TCONT_COUNTERS_T *pTcontCounters)**	
Description	Get the information about the specified TCONT.	
Support Chip	EN7521	
Parameter 1	aTcontCounters	The information of the specified TCONT
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.51 gponmgr_lib_get_driver_event_block

Function Prototype	**int gponmgr_lib_get_driver_event_block(XPON_EVENT_T *event)**	
Description	Get the block of driver events.	
Support Chip	EN7521	
Parameter 1	event	The block of driver event
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.52 gponmgr_lib_get_info

Function Prototype	**int gponmgr_lib_get_info(GPON_ONU_INFO_t * gponOnuInfo)**	
Description	Get the related information on the GPON ONU.	
Support Chip	EN7521	
Parameter 1	gponOnuInfo	the related information GPON ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.53 gponmgr_lib_get_phy_trans_params

Function Prototype	**int gponmgr_lib_get_phy_trans_params(PHY_PARAMS_t * pPhyTransParams)**	
Description	Get the parameters of the phy transceiver	
Support Chip	EN7521	
Parameter 1	pPhyTransParams	The parameters of the phy transceiver
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.54 gponmgr_lib_get_phy_fec_status

Function Prototype	**int gponmgr_lib_get_phy_fec_status(PHY_FECSTATUS_t *fecStatus)**	
Description	Get the status of phy FEC.	
Support Chip	EN7521	
Parameter 1	fecStatus	the status of phy FEC
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.55 gponmgr_lib_get_gemcounter

Function Prototype	**int gponmgr_lib_get_gemcounter(struct XMCS_GponGemCounter_S *pgponGemCounter)**	
Description	Get the information of the GEM counter.	
Support Chip	EN7521	
Parameter 1	pgponGemCounter	The information of GEM counter
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.56 xgponmgr_lib_get_dbg_level

Function Prototype	**int xgponmgr_lib_get_gemcounter(ushort *pgponDbgLevel)**	
Description	Get the XGPON MAC debug level.	
Support Chip	EN7521	
Parameter 1	pgponDbgLevel	used to storethe XGPON MAC debug level
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.57 xgponmgr_lib_get_o23_o4_ploam_ctrl

Function Prototype	**int xgponmgr_lib_get_o23_o4_ploam_ctrl (XGPON_SW_HW_SELECT *sel)**	
Description	Get the o3 and o4 ploam control status.	
Support Chip	EN7521	
Parameter 1	sel	used to store the o3 and o4 ploam control status
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.58 xgponmgr_lib_get_eqd_offset

Function Prototype	**int xgponmgr_lib_get_eqd_offset (struct XGMS_EqdOffset_S *pxgponEqdOffset)**	
Description	Get the Eqd detail information.	
Support Chip	EN7521	
Parameter 1	pxgponEqdOffset	used to store EqD detail information
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.59 xgponmgr_lib_get_info

Function Prototype	**int xgponmgr_lib_get_info(struct XGMS_XgponOnuInfo_S *pxgponOnuInfo)**	
Description	Get the xgpon detail information.	
Support Chip	EN7521	
Parameter 1	pxgponOnuInfo	used to store XGPON MAC info
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.60 xgponmgr_lib_XGPON_state

Function Prototype	**int xgponmgr_lib_XGPON_state (unsigned char *state)**	
Description	Get the xgpon state.	
Support Chip	EN7521	
Parameter 1	state	used to store the XGPON state
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.61 xgponmgr_lib_get_onuRespTime

Function Prototype	**int xgponmgr_lib_get_onuRespTime (unsigned short *onuRespTime)**	
Description	Get the xgpon onu response time.	
Support Chip	EN7521	
Parameter 1	onuRespTime	used to store the onu response time
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.62 xgponmgr_lib_get_sniffer_gtc

Function Prototype	**int xgponmgr_lib_get_sniffer_gtc (GPON_10G_DEV_SNIFFER_MODE_T *sniffer)**	
Description	Get the xgpon onu sniffer mode.	
Support Chip	EN7521	
Parameter 1	sniffer	used to store the onu sniffer mode.
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.63 xgponmgr_lib_get_ploamd_filter

Function Prototype	**int xgponmgr_lib_get_ploamd_filter (GPON_10G_DEV_PLOAMD_FILTER_MODE_T *filter)**	
Description	Get the ploamd msg(exclude profile) filter and profile ploam filter.	
Support Chip	EN7521	
Parameter 1	filter	used to store ploamd msg filter and profile ploam filter info
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.1.64 xgponmgr_lib_get_ploam_micerr_drop_mode

Function Prototype	**int xgponmgr_lib_get_ploam_micerr_drop_mode (XPON_Mode_t *mode)**	
Description	Get the ploam mic error drop mode.	
Support Chip	EN7521	
Parameter 1	mode	used to store ploam mic error drop mode
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.1.65 xgponmgr_lib_get_dying_gasp _mode

Function Prototype	**int xgponmgr_lib_get_dying_gasp _mode (GPON_10G_DYING_GASP_MODE_T *dyingGasp)**	
Description	Get the dying gasp mode.	
Support Chip	EN7521	
Parameter 1	dyingGasp	used to store dying gasp mode
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2 Set APIs

2.6.2.1 gponmgr_lib_set_omci_dbg_level

Function Prototype	**int gponmgr_lib_set_omci_debug_level(unsigned char DebugLevel)**	
Description	set the debug level of OMCI.	
Support Chip	EN7521	
Parameter 1	DebugLevel	debug level of OMCI, range 0~4
Return Value	return EPONMGR_SUCCESS/EPONMGR_FAIL	

2.6.2.2 gponmgr_lib_set_lan_debug

Function Prototype	**int gponmgr_lib_set_lan_debug(unsigned char LanDebug)**	
Description	Enable or disable the function of LAN debug .	
Support Chip	EN7521	
Parameter 1	LanDebug	LAN debug, whether enabled
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.3 gponmgr_lib_set_onu_version

Function Prototype	**int gponmgr_lib_set_onu_version(char *onu_ver)**	
Description	Set the version of the ONU as defined by the vendor.	
Support Chip	EN7521	
Parameter 1	Onu_ver	The version of ONU, the max length is 14 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.4 gponmgr_lib_set_vendor_id

Function Prototype	**int gponmgr_lib_set_vendor_id(char *VendorId)**	
Description	Set the vendor of the ONU.	
Support Chip	EN7521	
Parameter 1	VendorId	The vendor ID of ONU, 4 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.5 gponmgr_lib_set_equipment_id

Function Prototype	**int gponmgr_lib_set_equipment_id(char *equipmentId)**	
Description	Set the equipment ID of the ONU. It is used to identify the specific type of ONU.	
Support Chip	EN7521	
Parameter 1	equipmentId	The equipment ID of ONU, max length is 20 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.6 gponmgr_lib_set_operator_id_cap

Function Prototype	**int gponmgr_lib_set_operator_id_cap(char *operatorID_cap)**	
Description	Set the operator ID of the ONU capability ME.	
Support Chip	EN7521	
Parameter 1	operatorID_cap	the operator ID,max length is 4 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.7 gponmgr_lib_set_operator_id_auth

Function Prototype	**int gponmgr_lib_set_operator_id_auth(char * operatorID_auth)**	
Description	Set the operator ID of LOID authentication ME.	
Support Chip	EN7521	
Parameter 1	operatorID_auth	the operator ID of GPON LOID authentication, max length is 4
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.8 gponmgr_lib_set_loid

Function Prototype	**int gponmgr_lib_get_loid(char * Loid)**	
Description	Set the logical ONU ID. It is used to provide a way for the ONU to identify itself.	
Support Chip	EN7521	
Parameter 1	Loid	The LOID has a maximum, length of 24 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.9 gponmgr_lib_set_loid_password

Function Prototype	**int gponmgr_lib_set_loid_password(char * LOID_Password)**	
Description	Set a logical password. It is used to provide a way for the ONU to submit authentication credentials.	
Support Chip	EN7521	
Parameter 1	LOID_Password	The password of LOID, max length is 12 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.10 gponmgr_lib_set_enhanced_psk

Function Prototype	**int gponmgr_lib_set_enhanced_psk(unsigned char * enhanced_psk)**	
Description	Set the ONU enhanced PSK.	
Support Chip	EN7521	
Parameter 1	enhanced_psk	The enhanced PSK of ONU, 16 bytes
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.11 gponmgr_lib_set_link_mode

Function Prototype	**int gponmgr_lib_set_link_mode(char linkMode)**	
Description	Set the link mode of GPON.	
Support Chip	EN7521	
Parameter 1	linkMode	The link mode of GPON, 0:auto, 1:GPON, other:error
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.12 gponmgr_lib_set_vendor_pro_code

Function Prototype	**int gponmgr_lib_set_vendor_pro_code(unsigned short vendorProCode)**	
Description	Set the vendor specific product code for the ONU.	
Support Chip	EN7521	
Parameter 1	vendorProCode	vendor product code of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.13 gponmgr_lib_set_omcc_version

Function Prototype	**int gponmgr_lib_set_omcc_version(unsigned char omccVersion)**	
Description	Set the version of the OMCC protocol being used by the ONU.	
Support Chip	EN7521	
Parameter 1	omccVersion	The version of OMCC
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.14 gponmgr_lib_set_tm_option

Function Prototype	int gponmgr_lib_set_tm_option(unsigned char tmOption)	
Description	Set up the upstream traffic management function implemented in the ONU.	
Support Chip	EN7521	
Parameter 1	tmOption	The traffic management option of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.15 gponmgr_lib_set_survival_time

Function Prototype	int gponmgr_lib_set_survival_time(unsigned char onuSurvivalTime)	
Description	Set the minimum guaranteed time in milliseconds between the loss of external power and the silence of the ONU.	
Support Chip	EN7521	
Parameter 1	onuSurvivalTime	The survival time of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.16 gponmgr_lib_set_extended_tclayer

Function Prototype	int gponmgr_lib_set_extended_tclayer(unsigned char extendedTcLayer)	
Description	Set whether the ONU supports or does not support various optional TC-layer capabilities of [ITU-T G.984.3].	
Support Chip	EN7521	
Parameter 1	extendedTcLayer	Bit 1(LSB): ITU-T G.984.3 Annex C, PON-ID maintenance Bit 2: ITU-T G.984.3 Annex D, PLOAM channel enhancements: swift_POPUP and Ranging_adjustment messages. Bit 3..16: Reserved
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.17 gponmgr_lib_set_security_cap

Function Prototype	int gponmgr_lib_set_security_cap(unsigned char securityCap)	
Description	Set the security capability of the ONU.	
Support Chip	EN7521	
Parameter 1	securityCap	The security capability of the ONU 0 Reserved 1 AES-128 payload encryption is supported 2..255 Reserved
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.18 gponmgr_lib_set_total_ts_num

Function Prototype	int gponmgr_lib_set_total_ts_num(unsigned char totalTSNum)	
Description	Set the total number of traffic schedulers that are not associated with a circuit pack, but with the ONU in its entirety.	
Support Chip	EN7521	
Parameter 1	totalTSNum	The total traffic scheduler number of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.19 gponmgr_lib_set_total_pq_num

Function Prototype	int gponmgr_lib_set_total_ts_num(unsigned char totalPQNum)	
Description	Set the total number of upstream priority queues that are not associated with a circuit pack, but with the ONU in its entirety.	
Support Chip	EN7521	
Parameter 1	totalPQNum	total priority queue number for ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.20 gponmgr_lib_set_total_gemport_num

Function Prototype	int gponmgr_lib_set_total_gemport_num(unsigned char totalGEMPortNum)	
Description	Set the total number of GEM port-IDs supported by the ONU.	
Support Chip	EN7521	
Parameter 1	totalGEMPortNum	The total gemport number of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.21 gponmgr_lib_set_connectivity_cap

Function Prototype	int gponmgr_lib_set_connectivity_cap(unsigned short connectivityCap)	
Description	Set the Ethernet connectivity models that the ONU can support.	
Support Chip	EN7521	
Parameter 1	connectivityCap	the Ethernet connectivity models that the ONU can support
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.22 gponmgr_lib_set_qos_conf_flex

Function Prototype	int gponmgr_lib_set_qos_conf_flex(unsigned short qosConfFlex)	
Description	Set whether various MEs in the ONU are fixed by ONU's architecture or whether they are configurable.	
Support Chip	EN7521	
Parameter 1	qosConfFlex	QoS configuration flexibility
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.23 gponmgr_lib_set_voip_by_tr069

Function Prototype	**int gponmgr_lib_set_voip_by_tr069(char VoIPbyTR069)**	
Description	Setthe VoIP configuration method, whether is TR069.	
Support Chip	EN7521	
Parameter 1	VoIPbyTR069	Whether is TR069
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.24 gponmgr_lib_set_spec_version

Function Prototype	**int gponmgr_lib_set_spec_version(unsigned char SpecVersion)**	
Description	Set spec version of ONU capability ME.	
Support Chip	EN7521	
Parameter 1	SpecVersion	spec version of ONU capability ME
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.25 gponmgr_lib_set_onu_power_ctrl

Function Prototype	**int gponmgr_lib_set_onu_power_ctrl(unsigned char ONU_PowerCtr)**	
Description	Set whether the Tx power of the ONU can be controlled independently.	
Support Chip	EN7521	
Parameter 1	ONU_PowerCtr	0x00:do not support ONU power control 0x01:only supports ONU Tx power control 0x02: The ONU Tx and Rx power can be independently controlled
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.26 gponmgr_lib_set_auth_status

Function Prototype	**int gponmgr_lib_set_auth_status(unsigned char AuthStatus)**	
Description	Set the status of LOID authentication.	
Support Chip	EN7521	
Parameter 1	AuthStatus	The status of LOID authentication 0 Initial state, status indeterminate (requires further study) 1 Successful authentication 2 LOID error 3 Password error 4 Duplicate LOID
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.27 gponmgr_lib_set_max_ports_list

Function Prototype	**int gponmgr_lib_set_max_ports_list(char max_ports,int *port_id)**	
Description	Set the largest port number contained in the port list attributes.	
Support Chip	EN7521	
Parameter 1	max_ports	The max port number
Parameter 2	Port_id	Port number
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.28 gponmgr_lib_set_sr_indication

Function Prototype	**int gponmgr_lib_set_sr_indication (unsigned char srIndication)**	
Description	Set the queue status for the DBA.	
Support Chip	EN7521	
Parameter 1	srIndication	The value true means that status reporting is available for all T-CONTs that are associated with the ANI.
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.29 gponmgr_lib_set_total_tcont_num

Function Prototype	**int gponmgr_lib_set_total_tcont_num(unsigned short totalTcontNum)**	
Description	Set the total number of T-CONTs that can be supported on this ANI.	
Support Chip	EN7521	
Parameter 1	totalTcontNum	The total number of T-CONTs
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.30 gponmgr_lib_set_gem_block_length

Function Prototype	**int gponmgr_lib_set_gem_block_length(unsigned short gemBlockLength)**	
Description	Set the queue occupancy reporting granularity for the DBA, in units of bytes.	
Support Chip	EN7521	
Parameter 1	gemBlockLength	GEM block length
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.31 gponmgr_lib_set_onu_response_time

Function Prototype	**int gponmgr_lib_set_onu_response_time(unsigned short onuResponseTime)**	
Description	Set the ONU's actual response time.	
Support Chip	EN7521	
Parameter 1	onuResponseTime	The response time of the ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.32 gponmgr_lib_set_con_req_port

Function Prototype	**int gponmgr_lib_set_con_req_port(unsigned short conReqPort)**	
Description	Set the TCP or UDP port value as assigned by IANA for the management protocol associated with this virtual Ethernet interface.	
Support Chip	EN7521	
Parameter 1	conReqPort	IANA assigned port
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.33 gponmgr_lib_set_emergency

Function Prototype	**int gponmgr_lib_set_emergency (unsigned char EmergencyState)**	
Description	Set the emergency state of ONU.	
Support Chip	EN7521	
Parameter 1	EmergencyState	The emergency state of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.34 gponmgr_lib_set_qos_scheduler

Function Prototype	**int gponmgr_lib_set_qos_scheduler(struct XMCS_ChannelQoS_S *pwanChannelQos)**	
Description	Set the type of QoS scheduler.	
Support Chip	EN7521	
Parameter 2	pwanChannelQos	The type of QoS scheduler
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.35 gponmgr_lib_set_qos_weight_cfg

Function Prototype	**int gponmgr_lib_set_qos_weight_cfg (struct XMCS_QoSWeightConfig_S *pwanQosWeightCfg)**	
Description	Set the type of QoS weight.	
Support Chip	EN7521	
Parameter 1	pwanQosWeightCfg	The type of QoS weight 0:packet 1:byte
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.36 gponmgr_lib_set_congest_config

Function Prototype	**int gponmgr_lib_set_congest_config(unsigned int cmd, struct XMCS_TxQueueCongestion_S *pwanCongest)**
Description	Set the information of congestion.
Support Chip	EN7521

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Function Prototype	**int gponmgr_lib_set_congest_config(unsigned int cmd, struct XMCS_TxQueueCongestion_S *pwanCongest)**	
Parameter 1	cmd	Ioctl command
Parameter 2	pwanCongest	the information of congestion
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.37 gponmgr_lib_set_trtcm_scale

Function Prototype	**int gponmgr_lib_set_trtcm_scale (struct XMCS_TxTrtcmScale_S *pwanTrtcmScale, struct XMCS_TxQueueTrtcm_S *pwanTrtcmParameters)**	
Description	Set the scale of trTCM.	
Support Chip	EN7521	
Parameter 1	pwanTrtcmScale	The scale of trTCM
Parameter 2	pwanTrtcmParameters	The related parameters of trTCM
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.38 gponmgr_lib_set_trtcm_para

Function Prototype	**int gponmgr_lib_set_trtcm_para (struct XMCS_TxQueueTrtcm_S *pwanTrtcmParameters)**	
Description	Set the related parameters of trTCM.	
Support Chip	EN7521	
Parameter 1	pwanTrtcmParameters	The related parameters of trTCM
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.39 gponmgr_lib_set_tod_time

Function Prototype	**int gponmgr_lib_set_tod_time(struct XMCS_GponTodCfg_S *pgponTodCfg)**	
Description	Set ToD.	
Support Chip	EN7521	
Parameter 1	pgponTodCfg	ToD
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.40 gponmgr_lib_set_sn

Function Prototype	**int gponmgr_lib_set_sn(unsigned char *psn)**	
Description	Set the serial number of ONU.	
Support Chip	EN7521	
Parameter 1	psn	The serial number of ONU
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.41 gponmgr_lib_set_passwd

Function Prototype	**int gponmgr_lib_set_passwd(char *ppasswd)**	
Description	Set the password of the GPON serial number.	
Support Chip	EN7521	
Parameter 1	ppasswd	The password of the GPON serial number
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.42 gponmgr_lib_set_clear_gem_cnt

Function Prototype	**int gponmgr_lib_set_clear_gem_cnt(unsigned short gemportID)**	
Description	Set which GEM port counter to clear.	
Support Chip	EN7521	
Parameter 1	gemportID	which GEM port counter to clear
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.43 gponmgr_lib_set_clear_wan_cnt

Function Prototype	**int gponmgr_lib_set_clear_wan_cnt(WAN_TX_RX_SELECT_t mask)**	
Description	Clear the PPE Tx or Rx counter.	
Support Chip	EN7521	
Parameter 1	mask	Counter mask
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.44 gponmgr_lib_set_InitRepO1_Flag

Function Prototype	**int gponmgr_lib_set_InitRepO1_Flag(int value)**	
Description	Set the flag that specifies whether to report the O1 event in the initial process.	
Support Chip	EN7521	
Parameter 1	value	Flag value
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.45 gponmgr_lib_set_drv_rogueonu_mode

Function Prototype	**int gponmgr_lib_set_drv_rogueonu_mode(unsigned mode)**	
Description	Set whether to enable the rogue ONU.	
Support Chip	EN7521	
Parameter 1	mode	Mode
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.2.46 gponmgr_lib_set_dba_slight_modify

Function Prototype	int gponmgr_lib_set_dba_slight_modify(XPON_Mode_t enable)	
Description	Enable or disable DBA slight modify.	
Support Chip	EN7521	
Parameter 1	enable	
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.47 gponmgr_lib_set_dba_slight_modify_total

Function Prototype	int gponmgr_lib_set_dba_slight_modify_total(ushort total_size)	
Description	Set the total size of DBA backdoor.	
Support Chip	EN7521	
Parameter 1	size	Total size
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.2.48 gponmgr_lib_set_pon_link_state

Function Prototype	int gponmgr_lib_set_pon_link_state(XPON_Mode_t state)	
Description	Set the detection mode of the GPON WAN driver.	
Support Chip	EN7521	
Parameter 1	state	The detection mode of the GPON WAN driver, 0:Auto, 1:GPON and other:error
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3 Other APIs

2.6.3.1 gponmgr_lib_gemport_assign

Function Prototype	int gponmgr_lib_gemport_assign(int cmd, void *arg)	
Description	Assign the GEM port .	
Support Chip	EN7521	
Parameter 1	cmd	Ioctl command
Parameter 2	arg	Pointer of struct XMCS_GemPortAssign_S variable
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.2 gponmgr_lib_init

Function Prototype	int gponmgr_lib_init(void)
Description	GPONMGR initialization.
Support Chip	EN7521
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL

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2.6.3.3 gponmgr_lib_deinit

Function Prototype	**int gponmgr_lib_deinit(void)**
Description	Free GPONMGR memory space .
Support Chip	EN7521
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL

2.6.3.4 gponmgr_lib_init_omci_socket

Function Prototype	**int gponmgr_lib_init_omci_socket(int * socket_f)**	
Description	Initialize socket for receiving or sending an OMCI message.	
Support Chip	EN7521	
Parameter 1	socket_f	socket descriptor
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.5 gponmgr_lib_omci_receive_msg_block

Function Prototype	**int gponmgr_lib_omci_receive_msg_block(int socket,unsigned char * msg, unsigned int *msg_len)**	
Description	Receiving OMCI message.	
Support Chip	EN7521	
Parameter 1	socket	socket descriptor
Parameter 2	msg	OMCI message
Parameter 3	msg_len	The length of the OMCI message
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.6 gponmgr_lib_omci_receive_block

Function Prototype	**int gponmgr_lib_omci_receive_block(unsigned char *msg, unsigned int *msg_len)**	
Description	First call the gponmgr_lib_init_omci_socket function,then call gponmgr_lib_omci_receive_msg_block function.	
Support Chip	EN7521	
Parameter 1	msg	OMCI message
Parameter 2	msg_len	The length of the OMCI message
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.7 gponmgr_lib_omci_send_msg_block

Function Prototype	**int gponmgr_lib_omci_send_msg_block(int socket_f, unsigned char *msg, unsigned int msg_len)**
Description	Sending OMCI message.
Support Chip	EN7521

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Function Prototype	**int gponmgr_lib_omci_send_msg_block(int socket_f, unsigned char *msg, unsigned int msg_len)**	
Parameter 1	socket_f	socket descriptor
Parameter 2	msg	OMCI message
Parameter 3	msg_len	The length of the OMCI message
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.8 gponmgr_lib_omci_send_block

Function Prototype	**int gponmgr_lib_omci_send _block(int socket_f, unsigned char *msg, unsigned int msg_len)**	
Description	Call gponmgr_lib_omci_send_msg_block function.	
Support Chip	EN7521	
Parameter 1	msg	OMCI message
Parameter 2	msg_len	The length of the OMCI message
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.9 gponmgr_lib_bind_tcont_to_allocid

Function Prototype	**int gponmgr_lib_bind_tcont_to_allocid(unsigned short tcont, unsigned short allocid)**	
Description	Binding TCONT to alloc ID.	
Support Chip	EN7521	
Parameter 1	tcont	TCONT number
Parameter 2	allocID	Alloc ID
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.10 gponmgr_lib_unbind_tcont_to_allocid

Function Prototype	**int gponmgr_lib_unbind_tcont_to_allocid(unsigned short tcont)**	
Description	Unbinding TCONT and alloc ID.	
Support Chip	EN7521	
Parameter 1	tcont	TCONT number
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.11 gponmgr_lib_delete_tcont

Function Prototype	**int gponmgr_lib_delete_tcont(unsigned short tcont)**	
Description	Delete TCONT.	
Support Chip	EN7521	
Parameter 1	tcont	TCONT number
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

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2.6.3.12 gponmgr_lib_config_gemport

Function Prototype	**int gponmgr_lib_config_gemport(unsigned short portid, GPON_GemType_t type, unsigned short tcont)**	
Description	Create GEM port.	
Support Chip	EN7521	
Parameter 1	portid	GEM port ID
Parameter 2	type	0: unicast 1: multicast
Parameter 3	tcont	TCONT number
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.13 gponmgr_lib_delete_gemport

Function Prototype	**int gponmgr_lib_delete_gemport(unsigned short portid)**	
Description	Delete GEM port.	
Support Chip	EN7521	
Parameter 1	portid	GEM port ID
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL	

2.6.3.14 gponmgr_lib_reset_driver_data_service

Function Prototype	**int gponmgr_lib_reset_driver_data_service()**
Description	Delete all data services in the driver, leave only ONU ID and OMCC, stay on O5.
Support Chip	EN7521
Return Value	return ECONET_E_SUCCESS/ECONET_E_FAIL

2.7 Register map

List all registers (usually generated by CODA).

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